MILITARY SPECIFICATION

CRUSHER, JAW, PORTABLE, PRIMARY AGGREGATE PLANT, MOBILE, 75 TON PER HOUR CAPACITY

This specification is approved for use by the Naval Facilities Engineering Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a complete primary, portable, mobile, all electric powered plant capable of crushing aggregate at a production rate of 75 tons per hour (tph). The plant shall hereinafter be called the unit.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified (see 6.2), the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

L-P-378	- Plastic Sheet and Strip, Thin Gauge, Polyolefin.
QQ-S-781	- Strapping, Steel, and Seals.
TT-P-664	- Primer Coating, Synthetic, Rust-Inhibiting, Lacquer Resisting.
TT-V-119	- Varnish, Spar, Phenolic-Resin.
MMM-A-260	- Adhesive, Water-Resistant, (For Sealing Waterproofed Paper).
PPP-B-601	- Boxes, Wood, Cleated Plywood.
PPP-B-621	- Box, Wood, Nailed and Lock Corner.
PPP-B-1055	- Barrier Material, Waterproofed, Flexible.
PPP-P-40	- Packaging and Packing of Hand Tools.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer (Code 156), Naval Construction Battalion Center, Port Hueneme, CA 93043, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 3820

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MILITARY

MIL-C-104 - Crate, Wood, Lumber and Plywood Sheathed, Nailed and Bolted.
MIL-P-116 - Preservation, Methods of.
MIL-B-121 - Barrier Material, Greaseproofed, Waterproofed, Flexible.
MIL-V-173 - Varnish, Moisture-and Fungus-Resistant (For Treatment of Communications, Electronic, and Associated Equipment).
MIL-C-3774 - Crate, Wood, Open, 12,000 and 16,000 Pound Capacity.
MIL-C-5501 - Cap and Plug, Protective, Dust and Moisture Seal.
MIL-V-13811 - Varnish, Waterproofing, Electrical, Ignition.
MIL-E-16298 - Electric Machines Having Rotating Parts and Associated Repair Parts, Packaging of.
MIL-T-22085 - Tape, Adhesive, Preservation and Sealing.
MIL-B-22191 - Barrier Material, Transparent, Flexible, Heat Sealable.
MIL-P-46093 - Primer Coating, Synthetic (For Brake Drums).
MIL-C-52950 - Crate, Wood, Open, and Covered.

STANDARDS

FEDERAL

Fed. Std. No. 595 - Colors.

MILITARY

MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-209 - Slinging and Tiedown Provisions for Liftiing and Tying Down Military Equipment.
MS51336 - Lunette-Coupler, Drawbar, Ring.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents form a part of this specification to the extent specified herein.

DEPARTMENT OF LABOR (DoL)
OCCUPATIONAL SAFETY AND HEALTH ADMINISTRATION (OSHA)

Occupational Safety and Health Standards.

(Application for copies of DoL should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DODISS and the supplement thereto, if applicable.

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

C131 - Test for Resistance to Abrasion of Small Size Coarse Aggregate by Use of the Los Angeles Abrasion Machine.
C170 - Test for Compressive Strength of Natural Building Stone.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

AB1 - Molded Case Circuit Breakers.
ICS - Industrial Controls and Systems.
MG1 - Motor and Generators.
PB2 - Dead-Front Distribution Switchboards.

Application for copies should be addressed to the National Electrical Manufacturers Association, 2101 L Street, N.W., Washington, DC 20037.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

No. 70 - National Electrical Code.

(Application for copies should be addressed to the National Fire Protection Association, Batterymarch Park, Quincy, MA 02269.)

SOCIETY OF AUTOMOTIVE ENGINEERS, INC. (SAE)

SAE Handbook.

(Application for copies should be addressed to the Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale, PA 15096).

TIRE AND RIM ASSOCIATION, INC.

T&RA Yearbook.

(Application for copies should be addressed to the Tire and Rim Association, Inc., 3200 West Market Street, Akron, OH 44313.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Description. The unit shall consist essentially of a hopper, feeder jaw crusher, conveyer system, electric power system, operator's platform, and all accessories for a complete portable unit. The unit shall be mounted on a pneumatic tired semitrailer equipped with fifth wheel pin, applicable to any standard truck tractor, and a pneumatic tired towing front dolly.

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3.2 Standard commercial product. The unit shall, as a minimum, be in accordance with the requirements of this specification and shall be the manufacturer's standard commercial product. Additional or better features which are not specifically prohibited by this specification but which are a part of the manufacturer's standard commercial product, shall be included in the unit being furnished. A standard commercial product is a product which has been sold or is being currently offered for sale on the commercial market through advertisements or manufacturer's catalogs, or brochures, and represents the latest production model.

3.3 First article. When specified (see 6.2), the contractor shall furnish one complete unit for first article inspection and approval (see 4.2.1 and 6.4).

3.4 Materials. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components or of the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. Unless otherwise specified herein, all equipment, material, and articles incorporated in the work covered by this specification are to be new and fabricated using materials produced from recovered materials to the maximum extent possible without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products are allowed under this specification unless otherwise specified.

3.5 Interchangeability. All units of the same classification furnished with similar options under a specific contract shall be identical to the extent necessary to insure interchangeability of component parts, assemblies, accessories, and spare parts.

3.6 Maintainability. The unit shall be so designed and constructed that normal adjustment, repair, and overhaul can be readily accomplished by means of general purpose tools with a minimum removal or disturbance of other elements of the unit. Ease of maintenance provisions shall insure operating clearances for facilitating maintenance and servicing. Where possible, intricate locking devices, controls and threaded fastenings, that can be easily over-torqued by operators lacking feel through thick gloves or numbness, shall be avoided. Covers or plates that must be removed for component adjustment or for parts removal shall be equipped with substantial fastenings.

3.7 Safety. All rotating or reciprocating parts, and all parts subject to high operational temperature, that are of such nature or are so located as to be or become a hazard to operating or attending personnel, shall be adequately guarded or insulated to the extent necessary to eliminate the hazard. All platforms, steps, and handholds shall be provided for easy access to the operator's position and other work areas. The unit shall comply with OSHA regulations in effect at time of manufacture.

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3.8 Performance. The unit shall be capable of crushing any variety of rock having an average compressive strength of not less than 32,000 pounds per square inch (psi) when tested in accordance with ASTM C170, having a percentage of wear not exceeding 20 when tested in accordance with ASTM C131, and a free moisture content of not more than 6.50 percent. The unit shall be capable of crushing rock of all sizes including quarry rock up to 18 by 28 inches cross section and a length receptable to the primary jaw crusher opening depth (approximately 38 inches). The unit shall be capable of producing, with the crusher setting (close of stroke) set a 3, 4, 5, and 6 inches, the following minimum tph, 75, 100, 120, and 140 tons. Provisions shall be made for an optional belt reject conveyor to remove all or a portion of the fines.

3.9 Feeder. The unit shall be provided with an apron feeder, wobbler feeder, vibrating grizzly feeder, or a combination thereof as specified (see 6.2). The feeder shall be capable of delivering up to 180 tph. When the unit incorporates a wobbler feeder or vibrating grizzly feeder the bar spacing shall be adjustable from not less than 3 to 6 inches. Replaceable liners, flights, bars, or plates shall be fabricated of not less than 11 percent maganese alloy steel or high quality abrasive-resisting steel plate. Housing end plate minimal thickness shall be 1.250 inches Deck plate minimal thickness shall be 0.750 inch. When feeder incorporates a wobbler or vibrating grizzly the feeder shall include a by-pass spout and reject spout with gate to the reject conveyor. Protection shall be provided against shock and vibration being transmitted to the unit structures The feeder shall be controlled from the operator's platform.

3.10 Loading hopper. A loading hopper extending not less than the full length of the feeder with a vertical height above the feeder pan of not less than 24 inches shall be provided. The loading hopper shall break down into three sections (two sides and a rear). The feeder or feeder support and hopper shall be removable to reduce cube during transport. The sides and rear shall be mounted at an angle of not less than 30° from the horizontal. The loading hopper shall be made of not less than 0.750-inch plate with supports and brackets for wider extension. The top edge of the loading hopper shall be structurally reinforced to receive and support an additional wobbler or grizzly if required at a later date.

3.11 By-pass chute. A by-pass chute shall be provided to allow material smaller than the closed jaw setting to by-pass the jaw crusher. The chute shall be located between the feeder and the jaw crusher. The wobbler or grizzly shall be mounted across the chute opening to route material into the jaw crusher. The chute shall be protected at the point of impact of material by suitable replaceable liners with access for replacement.

3.12 Jaw crusher. The jaw crusher for the unit shall be V-belt driven from the unit motive power, and shall be of the overhead eccentric single jaw type with a jaw size opening of not less than 20 by 36 inches. The jaws shall be approximately 40 inches in depth with the lower jaw opening adjustable from approximately 2 to 6 inches in the closed position. The jaw crusher shall be of the hydraulic-shim adjusted type with reversible stationary and removable

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jaw plates, with self-aligning antifriction or tapered roller bearings. Cheek and jaw plates shall be of not less than 11 percent manganese content. Jaw plates shall be equipped with toggle plates or safety relief plates for the prevention of overload damage by noncrushable material (tramp iron). The eccentric shaft shall be fitted with counterweighted fly-wheels with sufficient mass to maintain relatively constant rotational speed throughout each revolution when crushing rock of maximum size and hardness. An extra set of cheek and jaw plates with appropriate toggle plates shall be furnished with the unit.

3.13 Conveyors. Conveyors shall be used to transport material within the unit and shall be of the belted type, in accordance with the manufacturer's standard practice, providing a smooth and efficient flow of material. A conveyor shall be provided to transfer material from the jaw crusher and by-pass chute to the main product conveyor. The main product conveyor shall be capable of delivering 180 tph, discharging the product not less than 8 feet from the trailer frame and not less than 8 feet above ground level. A conveyor shall be provided for transferring material from the reject spout.

3.13.1 Belt-type conveyors. Belt-type conveyors shall be equipped with troughing type idlers or rollers with spacing not exceeding 4 feet between centers. Troughing idlers shall be so designed as to insure true running and self-centering of the conveyor belts. Special rolls designed to absorb shock shall be installed under the conveyor belt, below receiving area. Return idlers shall be provided with a maximum spacing of 10 feet between centers. All idlers shall be at least 4 inches in diameter. The width and running speed of all conveyor belts shall be such as to assure efficient movement of material, with a minimum of spillage and pileup at receiving points, when the unit is operating at full capacity. An accessible adjustment for maintaining proper belt tension shall be provided. Cleaning devices, capable of removing foreign material adhering to the pulley side of the belt, shall be provided. Conveyors which extend outside the basic dimensional limits of the unit shall be of the folding, retractable, or readily demountable type. Rubber coating on the carrying side of the belt shall be not less than 3/16-inch thick and rubber coating on the bottom side shall be not less than 1/16-inch thick. Each belt-type conveyor shall be driven from a separate electric or hydraulic motor. The capacity of the motors shall be sufficient to handle the loads encountered when the plant is operating at maximum capacity. Motors shall be positioned to transmit power directly or by V-belt to the head pulley. Inclined conveyors shall be designed to prevent rollback of the belt in the event of power failure.

3.14 Operator's platform. The operator's platform shall be centrally located to enable the operator to observe the units feeder, conveyor, and crusher operation. Controls for plant operation shall be mounted on the platform for safe convenient operation of the unit by the operator. Access ladders, handrails, and guardrails shall be provided. Antiskid surfaces shall be provided for the deck, catwalks, and ladders. Platforms and catwalks shall be furnished to provide access to points requiring routine maintenance and adjustment. All guardrails around platform shall be demountable to reduce cube during transport.

3.15 Bearings. All antifriction or bronze bearings used on the unit, exposed to the elements, shall be provided with a dust-sealed device to prevent the entry of abrasive materials.

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3.16 Electrical equipment. All electrical motors shall conform to NEMA MG1 polyphase induction, totally enclosed, fan-cooled type. The rated horsepower (hp) of each motor shall be equal to, or greater than, the hp requirements of the equipment driven under all operating conditions. The hp of the motors shall be such as to allow progressive starting of all electrically operated components. The electric motors shall be suitable for operation on 460 volt (V), 60 Hertz (Hz), 3 phase, alternating current from an outside source. A main distribution panel conforming to NEMA PB2 shall provide branch circuit breakers for each electrical branch circuit, and a main circuit breaker that will disconnect the entire unit from the electric power supply. Circuit breakers shall conform to NEMA AB1. The main distribution panel shall be provided with a main feeder power receptacle, auxiliary main feeder outlet receptacle, and five auxiliary branch circuit receptacles mounted in a bank for receiving connections from portable conveyors. All power receptacles shall have spring-loaded covers. A motor controller operated by momentary contact start, and maintained contact stop pushbuttons shall be provided for control of each electric motor. Motor controllers shall conform to NEMA ICS. Motor control pushbuttons and an emergency panic pushbutton, which controls tripping of the master circuit breaker, shall be installed in an easily accessible location at the operator's platform and each control station for each electric motor used on the unit. Panic pushbutton enclosure shall be red. All enclosures for motors, motor controllers, distribution panel, pushbuttons, and other similar electrical apparatus shall be weatherproof and dustproof in accordance with NEMA ICS - 6, type 4 enclosures. All wiring of the unit shall be furnished, including; a 100-foot main feeder cable, interconnections between individual components and between components and the distribution panel, to provide an operable unit when connected to a 460 V, 60 Hz, 3 phase power source. A permanently legible wiring diagram shall be installed on or near the main distribution panel. The function of all circuit breakers and controls shall be permanently marked on their respective panels or enclosures. Conductors shall be color coded or suitably identified. All wiring shall be adequately protected from physical damage. The electrical system, wiring practices, system protection, grounding, and wiring hardware shall meet the applicable requirements of NFPA No. 70.

3.17 Semitrailer. The maximum stress imposed on any structural component of the semitrailer shall not exceed 60 percent of the yield point. The semitrailer for the crushing unit shall be equipped with a dolly and shall be capable of being safely towed over the road at speeds of at least 15 miles per hour (mph). The dolly drawbar shall be equipped with a lunette ring in accordance with MS51336. The angle of departure shall be at least 25° and projections below the main frame members shall be held to a minimum. The gross weight on any one axle, with the unit prepared for travel, shall not exceed 20,000 pounds. The semitrailer shall be equipped with straight air service brakes controllable from the towing vehicle cab. All necessary air hose and standard air couplings for connecting to the brake system of a towing vehicle shall be provided. The service brake installation shall include a breakaway system capable of automatically applying the brakes in case the semitrailer is accidentally separated from the towing vehicle, and shall be capable of maintaining the application for at least 15 minutes. Parking brakes, or chock blocks securely chained to the vehicle frame, shall be

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provided for each wheel. Chock block chains shall be of sufficient length to permit placing the blocks either ahead or behind the wheels. The blocks shall be equipped with snaphooks for storage. At least six screw-jacks shall be provided with each unit. The jacks shall be capable of supporting the unit when fully equipped and fully loaded. Supports at each end of the unit and near the crusher jaw shall be provided for blocking during unit operation.

3.17.1 Tires and rims. The tire and rim ratings shall conform to T&RA recommendations for the type and size of tires furnished. Tire and rim sizes shall be the same for all wheels. Tires shall be tube or tubeless type with highway tread. Tires shall be of rated capacity at least equal to the load imposed on each tire, measured at each wheel at the ground. Tires shall be not less than 100 level quality and shall be of domestic make. When tube type tires are furnished, inner tubes shall be heavy-duty type, and shall be of proper size for the tires furnished.

3.18 Identification plate. The contracting officer will furnish to the Government inspector the required identification plates. The contractor will be required to stamp the necessary data in the blank spaces thereon and securely affix said plates in a conspicuous place on each unit, assembly or subassembly, and parts as directed by the Government inspector. Nonferrous screws, rivets, or bolts of not less than 1/8 inch in diameter shall be used to affix the plates. The applicable nomenclature contained in the contract item description shall be placed in the top blank.

3.19 Cleaning, treatment, and painting. Surfaces normally painted in good commercial practice shall be cleaned, treated, and painted as specified herein. The color of the finish coat shall be green, No. 14064 conforming to Fed. Std. No. 595. Surfaces to be painted shall be cleaned and dried to insure that they are free from contaminants such as oil, grease, welding slag and spatter, loose mill scale, water, dirt, corrosion product, or any other contaminating substances. As soon as practicable after cleaning, and before any corrosion product or other contamination can result, the surfaces shall be prepared or treated to insure the adhesion of the coating system. The painting shall consist of at least one coat of primer and one finish coat. The primer shall be applied to a clean, dry surface as soon as practicable after cleaning and treating. Painting shall be with manufacturer's current materials according to manufacturer's current processes and the total dry film thickness shall be not less than 2.5 mils over the entire surface. The paint shall be free from runs, sags, orange peel, or other defects.

3.20 Instruction plates. The unit shall be equipped with instruction plates suitably located, describing any special or important procedures to be followed in operating and servicing the equipment. Plates shall be of a material which will last and remain legible for the life of the equipment, and shall be securely affixed thereto with nonferrous screws or bolts of not less than 1/8-inch diameter.

3.21 Toolbox(s). One or more weatherproof toolbox(s) shall be provided. The toolbox(s) shall be large enough to store all tools required for field service or maintenance. The toolbox(s) shall be steel with a hinged lid and a trunk drawbolt or heavy-duty draw pull catches to keep the lid secure when unit is subjected to vibration. The toolbox(s) shall be mounted in a protected accessible location.

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3.22 Stenciling. The gross vehicle weight of each demountable component of the unit shall be stenciled on each side of the component in such a manner as to be readily discernible to personnel handling the equipment. The total gross weight of the unit shall be stamped on the identification plate by the contractor.

3.23 Prime equipment accessories. Prime equipment shall be shipped with all required accessories and tools unless written deviation is received from the contracting officer.

3.24 Lifting and tying down attachments. The unit shall be equipped with lifting and tying down attachments. Lifting and tying down attachments shall conform to MIL-STD-209, type III. Attachments shall be indicated by a transportation plate except tying down attachments may be identified by stenciling or other suitable marking on the unit. Tying down markings shall clearly indicate that the attachments are intended for tying down of the unit on the carrier when shipped. A diagram showing the lifting attachments and lifting sling shall be inscribed on the transportation plate, with the lifting capacity of each attachment and the required length and size of each sling cable marked thereon. A silhouette of the crusher showing the center of gravity shall be provided.

3.25 Lubrication. Lubrication shall be in accordance with the manufacturer's standard practice. The lubricating points shall be easily visible and accessible. All parts requiring lubrication shall be lubricated as specified in section 5. Hydraulic lubrication fittings shall be in accordance with SAE J534. Where use of high pressure lubricating equipment, 1,000 psi or higher will damage grease seals or other parts, a suitable warning shall be affixed to the equipment in a conspicuous location.

3.26 Fungus resistance. When specified (see 6.2), electrical components and circuit elements, including terminal and circuit connections, shall be coated with varnish conforming to MIL-V-173, except that:

a. Components and elements inherently inert to fungi or in hermetically sealed enclosures need not be coated.
b. Current-carrying contact surfaces, such as relay contact points, shall not be coated.

3.27 Workmanship.

3.27.1 Steel fabrication. The steel used in fabrication shall be free from kinks, sharp bends, and other conditions which would be deleterious to the finished product. Manufacturing processes shall not reduce the strength of the steel to a value less than intended by the design. Manufacturing processes shall be done neatly and accurately. All bends shall be made by controlled means to insure uniformity of size and shape.

3.27.2 Bolted connections. Bolt holes shall be accurately punched or drilled and shall have the burrs removed. Washers or lockwashers shall be provided in accordance with good commercial practice, and all bolts, nuts, and screws shall be tight.

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3.27.3 Riveted connections. Rivet holes shall be accurately punched or drilled and shall have the burrs removed. Rivets shall be driven with pressure tools and shall completely fill the holes. Rivet heads, when not countersunk or flattened, shall be of approved shape and of uniform size for the same diameter of rivet. Rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member.

3.27.4 Welding. Welding procedures shall be in accordance with a nationally recognized welding code. The surface of parts to be welded shall be free from rust, scale, paint, grease, or other foreign matter. Welds shall be of sufficient size and shape to develop the full strength of the parts connected by the welds. Welds shall transmit stress without permanent deformation or failure when the parts connected by the weld are subjected to proof and service loadings.

3.27.5 Castings. All castings shall be sound and free from patching, misplaced coring, warping, or any other defect which reduces the castings ability to perform its intended function.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Material inspection. The contractor is responsible for insuring that supplies and materials are inspected for compliance with all the requirements specified herein and in applicable referenced documents.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2.1).
b. Quality conformance inspection (see 4.2.2).

4.2.1 First article inspection. The first article inspection shall be performed on one complete unit when a first article is required (see 3.3 and 6.2). This inspection shall include the examination of 4.3 and the tests of 4.4. The first article may be either a first production unit or a standard production unit from the supplier's current inventory provided the item meets the requirements of the specification and is representative of the design, construction, and manufacturing technique applicable to the remaining items to be furnished under the contract.

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4.2.2 Quality conformance inspection. The quality conformance inspection shall include the examination of 4.3, the tests of 4.4.2, and the packaging inspection of 4.5.

4.3 Examination. Each unit shall be examined for compliance with the requirements specified in section 3 of this specification. Any redesign or modification of the contractor's standard product to comply with specified requirements, or any necessary redesign or modification following failure to meet specified requirements shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual examinations and dimensional measurements. Noncompliance with any specified requirements or presence of one or more defects preventing or lessening maximum efficiency shall constitute cause for rejection.

4.4 Tests. The first article unit shall be tested as specified in 4.4.1, and each production unit shall be subjected to the operational test specified in 4.4.2.

4.4.1 First article tests. The first article unit shall be subjected to the tests specified in 4.4.1.1 through 4.4.1.8, and 4.4.2. Failure to pass one or more tests shall be cause for rejection.

4.4.1.1 Performance test. The crusher shall be tested as a unit as specified in 3.1. Quarry rock shall be of the material as specified in 3.8 containing some rock 18 inches by 28 inches. Only that maintenance established by the contractor and submitted as a maintenance schedule prior to the tests shall be performed during the test.

4.4.1.2 Percentage of crush. Arrange the flow of material so that all material passing through the wobbler or vibrating grizzly bars will be discharged to the stub conveyor or chute. Weigh not less than 100 tons of crushed material discharged from the main product conveyor. The material bypassed through the wobbler or grizzly bars shall be weighed separately and shall not exceed 20 percent of the combined total of material from the discharge chute and the main product conveyor. Presence in the feed of more than 20 percent of material not requiring crushing shall constitute failure of this test.

4.4.1.3 Operating time. Crushed rock production capacity shall be computed on the basis of net operating time which shall be defined as gross operating time less allowable down time for the following:

a. Jamming from rock fragments larger than the maximum size of rock specified in 3.8.
b. Delay due to lack of material delivery to the crusher.
c. Normal machinery maintenance: lubrication, jaw setting adjustments, and takeups (down time due to any design deficiency shall not be considered normal maintenance).
d. Other conditions considered by the contracting officer to be beyond the control of the contractor.

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Down time required for maintenance other than lubrication and operating adjustments shall be considered as net operating time. Down time caused by design deficiencies or failure of parts not classified as high mortality parts shall constitute failure of any test.

4.4.1.4 Jaw settings. When the crusher is tested as a unit, the jaw settings shall be as follows:

Closed setting in inches	Time in hours
Maximum (not less than 5 inches)	10
Minimum (not more than 3 inches)	10

Capacity in tph and corresponding jaw setting shall be recorded as part of test reports.

4.4.1.5 Capacity as a unit. Operate the unit under the applicable test conditions specified in 4.4.1.1 through 4.4.1.4. Cumulative daily runs are permissible. Nonconformance to 3.8 shall constitute failure of this test.

4.4.1.6 Uncrushable material test. With the unit operating at capacity rate, drop a piece of uncrushable material such as tramp iron, into the crusher. The piece shall be larger than the maximum jaw opening at the jaw setting being used, but not more than 2 inches larger. Nonconformance to 3.12 shall constitute failure of this test.

4.4.1.7 Holdback device test. During performance tests cut off power to motor-driven conveyors when carrying capacity load to determine conformance to 3.13.1. Failure to prevent slippage or reversal shall constitute failure of this test.

4.4.1.8 Lifting and tying down attachment test. When furnished, lifting and tying down attachments shall be tested to determine conformance to 3.24. The manufacturer may furnish certified engineering calculations in lieu of a test to prove that the lifting and tying down attachments offered meets the requirements set forth in MIL-STD-209

4.4.2 Production unit operational test. Each unit produced in fulfillment of a Government contract or order shall be completely assembled, adjusted, lubricated, or otherwise serviced for operation. The motors shall be started and subjected to a warmup period as recommended by the manufacturer. The unit shall be given a run-in test and all controls operated a sufficient number of times to ascertain that the components and mechanisms actuated by the controls operated promptly, fully, and without restriction or malfunction. Each wheeled unit shall be towed a distance of not less than 10 miles at an average speed of not less than 15 mph over unpaved road. Failure to pass any phase of this test shall be cause for rejection.

4.5 Packaging inspection. The preservation, packing, and marking of the unit shall be inspected to verify conformance to the requirements of section 5.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or commercial as specified (see 6.2).

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5.1.1 Level A.

5.1.1.1 Methods of preservation. Cleaning processes, drying procedures, preservatives, and methods of preservation are listed in MIL-P-116 and shall conform to the requirements of MIL-P-116 and any applicable specifications.

5.1.1.2 Disassembly. Disassembly shall be the minimum necessary to protect parts subject to damage or loss, and to accomplish reduction in cube. Removed bolts, nuts, pins, screws and washers shall be reinstalled in mating parts and secured to prevent their loss.

5.1.1.3 Matchmarking. Parts removed and mating parts on the equipment and attachments shall be matchmarked to facilitate reassembly. Parts and accessories removed, and mating parts on the equipment, shall be identified with weatherproof tags attached to mating parts and locations. Markings shall be applied to the tags with a waterproof material.

5.1.1.4 Cleaning and drying. Prior to the application of preservative compounds or paint, surfaces shall be cleaned by process C-1 and dried by any applicable procedure of MIL-P-116.

5.1.1.5 Unprotected surfaces. Unprotected exterior metal surfaces requiring the application of a contact preservative in accordance with MIL-P-116 and not specifically provided for herein shall be preserved as follows:

5.1.1.6 Unfinished (not machined) surfaces. Unfinished exterior metal surfaces shall be coated with type P-1 preservative.

5.1.1.7 Machined surfaces. Exposed machined surfaces shall be coated with type P-6 or P-11 preservative and wrapped or covered, as applicable, with barrier material conforming to MIL-B-121, type I, grade A, class 2 or MIL-B-22191, type II. The material shall be secured in place with waterproof tape.

5.1.1.8 Drive belts and pulleys. Drive belts shall be removed or released from tension. Removed belts shall be preserved method IC-1 or IC-3. Unpainted surfaces shall be coated with primer conforming to TT-P-664 or MIL-P-46093. A weatherproof tag shall be attached in a conspicuous location indicating: "Belts have been (removed or released from tension.) (Install or tension) prior to operation." Markings shall be applied to the tags with a waterproof material.

5.1.1.9 Electrical motors. Preserve level A by the alternate method of MIL-E-16298.

5.1.1.10 Electrical modules. Circuitry modules, printed circuit boards, and panel boards that are not hermetically sealed should be removed, if possible, and packaged method II. If the complexity of the circuitry will not allow removal, seal area with a waterproof barrier conforming to L-P-378 or MIL-B-22191, type III, or tape conforming to MIL-T-22085.

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5.1.1.11 Conveyor belts. Metal lacings shall be coated with varnish conforming to TT-V-119 or MIL-V-13811. Each belt shall be completely rolled with a tube or piece of round wood placed in each end fold of the belt to prevent sharp bends. Prior to rolling, belt surfaces shall be dusted with powdered talc to prevent surfaces from adhering together. When required, to hold belt in compact roll, strapping conforming to QQ-S-781, class 1, type I, finish B, shall be placed around the roll. Suitable protective cushioning shall be placed under the strapping, and the strapping not tensioned so tight as to damage the belting. Each rolled belt shall be wrapped with waterproofed barrier material conforming to PPP-B-1055, class E-1 or E-2, with all laps, seams, and folds sealed with adhesive conforming to MMM-A-260.

5.1.1.12 Brakes.

5.1.1.12.1 General. Clean all surfaces as required to remove contaminants. Coat unpainted exterior surfaces of rods, levers, cables, clevises, linkage, and associated parts with type P-1 preservative. Remove brake drum covers and spray enclosed metallic components with primer TT-P-664 or MIL-P-46093. Spray coat brake drum friction surfaces with a very thin coat (0.5-0.75 mil thick) of primer. Coat friction surfaces of cams, adjusting screws, and anchor pins with type P-6 or P-7 preservative. Backoff all brake shoe adjusters for maximum clearance between brake drum and brake shoes. Do not apply coating of any type to contact surfaces of electric brakes. Adjust brake shoes to provide maximum clearance to drum surfaces and lined surfaces. In a conspicuous location in the operator's compartment, attach a weatherproof warning tag marked, "Brakes preserved. Do not apply brakes when vehicle is being moved. Use tow bar or similar arrangement for moving vehicle. Adjust brakes before placing equipment into operation." Markings shall be applied with waterproof material.

5.1.1.12.2 Air-actuated brake system. The air-actuated brake systems shall be drained. Interior surfaces of the air-supply tanks shall be fogged with type P-9 or P-10 preservative. Excess preservative oil shall be drained. Threaded openings and the threads of drain plugs shall be coated with P-10. The drain plugs shall be reinstalled. Air line filters shall be drained and closed. The exhaust ports of relay emergency, quick-release, and relay valves not equipped with exhaust check valves shall be closed by inserting caps or plugs as applicable conforming to MIL-C-5501 or by sealing the ports with waterproof tape. Attach a waterproof tag to the valves and service lines indicating, "Remove (plugs, caps, or tape) from exhaust ports and valves before operating." Markings shall be applied with a waterproof material.

5.1.1.13 Lubrication. Lubricate all grease fittings with type P-11 preservative. Wheel bearings should be lubricated with a compound specified by the manufacturer. Lubricate control mechanisms, linkage, hinges, fasteners, and all pivot points with P-9.

5.1.1.14 Rubber tires. Tires mounted on the wheels (road tires) of the unit shall be inflated to 10 pounds above pressure value recommended for maximum load.

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5.1.1.15 Maintenance tools. Maintenance tools shall be preserved in accordance with level A preservation and packaging requirements of PPP-P-40.

5.1.1.16 Technical publications. Technical publications for each piece of equipment shall be preserved method IC-1 or IC-3.

5.1.1.17 Consolidation. Only components of one unit shall be consolidated together. Tool boxes shall be utilized for the consolidation of loose components, tools, and publications. The remainder of the components that will not fit in the tool boxes shall be consolidated, along with disassembled components, in boxes conforming to PPP-B-601, overseas type or PPP-B-621, class 2. Contents shall be cushioned, blocked, and braced to prevent movement within the containers. Containers shall be secured to the equipment with appropriate strapping. Arrangement and location on the equipment shall be such so as not to increase cubage or interfere with lifting or mobility of the equipment.

5.1.2 Commercial. The unit shall be preserved in accordance with the contractor's standard practice in a manner to prevent deterioration and damage.

5.2 Packing. Packing shall be level A or commercial as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Basic unit (except conveyors). The basic unit shall be prepared for mobile shipment, openings shall be sealed to prevent the entry of free falling water. Large disassembled components that cannot be consolidated, packaged, or secured to the basic unit shall be packed in crates conforming to MIL-C-52950 or MIL-C-3774. The contents shall be anchored, blocked, and braced in accordance with the appendix to the crate specification.

5.2.1.2 Portable conveyors. Portable conveyors shall be shipped knocked down to the most practicable extent. The structural components shall be nested, arranged, and secured with bolts or steel strapping, in combination with suitable wood blocking or battens, as required to form nonshifting bundles. The strapping shall be 0.035 by 1-1/4 inch strapping conforming to QQ-S-781, class 1, type I, finish B. The strapping shall be stapled to any wood blocking or battens provided. Where it is not practicable to disassemble the equipment to the extent required for suitable bundling, the sections shall be secured to skid bases conforming to the skid type base of MIL-C-104, style A crates, and shipped as individual assemblies. The electric motors, conveyor belts, bolts, nuts, washers, and any other small components requiring the protection of a container for safe convenient handling shall be packed in containers as specified for disassembled components. Where skid bases are used, consideration shall be given to consolidating and securing any bundled or boxed components on the skid bases.

5.2.2 Commercial. The unit shall be prepared for shipment in a manner which will insure arrival at destination in a satisfactory condition. Preparation for delivery shall comply with applicable carrier rules and regulations.

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5.3 Marking. Marking shall be in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The jaw crusher is intended for use as the primary rock-crushing unit in a rock crushing plant.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Issue of document required if other than as specified (see 2.1.1).
c. When a first article is required for inspection and approval (see 3.3, 4.2.1, and 6.4)
d. Type of feeder required (see 3.9).
e. When fungus resistance is required (see 3.26).
f. Level of preservation and level of packing required (see 5.1 and 5.2).

6.3 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423 and invokes the provisions of paragraph 7-104.9(n) of the Defense Acquisition Regulations (DAR), the data requirements will be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the Contract Data Requirements List (DD Form 1423) incorporated into the contract. When the provisions of DAR 7-104.9(n) are not invoked, the data shall be delivered in accordance with the contract requirements.

6.4 First article. When a first article inspection is required, the unit will be tested and should be a first production unit or it may be a standard production unit from the contractor's current inventory as specified in 4.2.1. The first article should consist of one unit. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.

Preparing activity:

Navy - YD

Project No. 3820-N010

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